Telco Churn

MSITM 6341 - Group A midterm project

- 1. Introduction to the Dataset
- 2. Data Cleaning and Preprocessing Steps
- 3. EDA & Key Insights
- 4. Model Building and Evaluation
- 5. Model implementation
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Data: IBM Telco Customer Churn (Kaggle).
Models: Logistic Regression and Random Forest (scikit-learn).
Authors: Richard Hansel - Katy Huang - Angela Cortes

Telco Customer Churn

Source: Kaggle - Telco Customer Churn (BlastChar), an IBM sample dataset of a fictional telecom company. Each row is one customer; **Churn** says whether they left in the last month.

Customers (rows)	Columns	Churn rate	Missing values found
7,043	21	26.5%	11

Variables

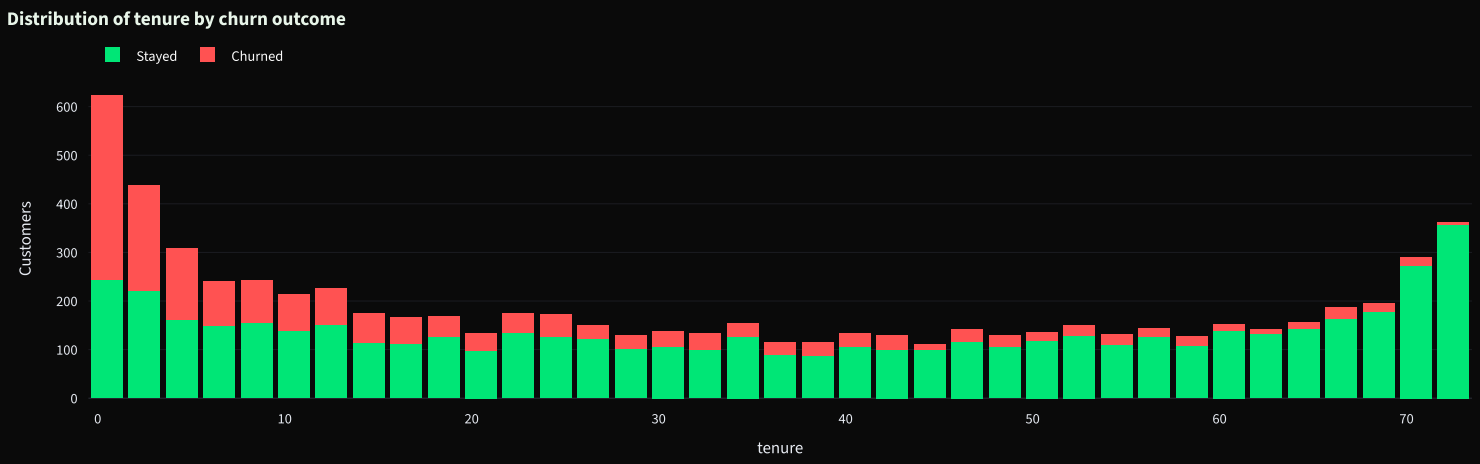
Built from the cleaned data, so TotalCharges is numeric and SeniorCitizen reads Yes/No.

Variable	Type	Range / categories	Description
customerID	Categorical	7,043 unique values	Unique customer identifier (not used for modeling)
gender	Categorical	Female, Male	Customer gender
SeniorCitizen	Categorical	No, Yes	Whether the customer is 65 or older
Partner	Categorical	No, Yes	Whether the customer has a partner
Dependents	Categorical	No, Yes	Whether the customer has dependents (children, etc.)
tenure	Numeric	0.00 to 72.00	Number of months the customer has been with the company
PhoneService	Categorical	No, Yes	Has a phone line
MultipleLines	Categorical	No, Yes	Has more than one phone line
InternetService	Categorical	DSL, Fiber optic, No	Type of internet connection
OnlineSecurity	Categorical	No, Yes	Subscribes to the online security add-on

Explore a variable

Variable

tenure


Distribution of tenure by churn outcome
Stayed
Churned
Customers
600
500
400
300
200
100
0
0
10
20
30
40
50
60
70
tenure

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Cleaning and Preprocessing

Every step below comes from `src/task_2_1_clean_preprocess.py` . The decision notes are read directly from the code's docstrings.

Before and after

Raw data	Cleaned data	Model-ready (train + test)
7,043 x 21	7,043 x 21	5,634 + 1,409 x 28

Cleaning

1. Strip whitespace

`strip_whitespace()` - b1. Strip leading/trailing spaces from every text column.

2. Fix TotalCharges type and impute blanks

`handle_missing_values()` - b3. Report missing values and impute blank TotalCharges with 0.

Blank TotalCharges	Of those with tenure 0	Imputed value
11	11 of 11	$0

> **DECISION:** impute TotalCharges = 0 rather than drop or use the mean/median. Every one of these customers has tenure == 0: they just signed up and have not been billed yet, so their true total really is $0. Imputing 0 records that true value and keeps all 7,043 customers. Dropping 11 rows (0.16%) would also be acceptable but discards valid customers, and mean/median imputation would wrongly give brand-new customers thousands of dollars in charges.

3. Remove duplicates

`remove_duplicates()` - b4. Report and remove duplicate rows and duplicate customer IDs.

Duplicate rows	Duplicate customer IDs
0	0

4. Check charges consistency

`check_charges_consistency()` - b5b. Flag customers whose TotalCharges doesn't match tenure x MonthlyCharges.

5. Standardize categories

`standardize_categories()` - b6. Make category labels consistent and remove redundant values.

"No internet service" → "No"	"No phone service" → "No"	SeniorCitizen

9,156 values | 682 values | 0/1 → No/Yes

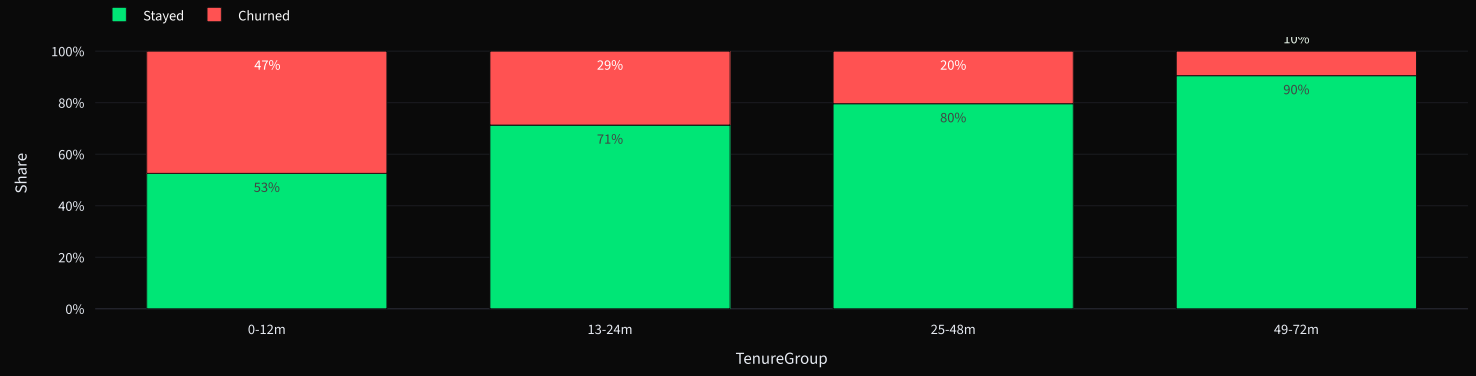
> **DECISION:** recode SeniorCitizen from 0/1 to "Yes"/"No". Every other yes/no column uses text labels, so this keeps the cleaned EDA file consistent and makes charts readable. It is encoded back to 0/1 in preprocessing.

> **DECISION:** recode "No internet service" (six add-on columns) and "No phone service" (MultipleLines) to "No". That information is already captured by the InternetService and PhoneService columns, so keeping it would only create redundant dummy columns later. "No" is accurate: the customer does not have that add-on.

Engineered features

- **NumServices:** how many add-on services the customer pays for (0-8). Customers who are more "embedded" in the product may be less likely to leave.
- **AvgMonthlySpend:** TotalCharges / tenure, i.e. what they actually paid per month on average. Differs from MonthlyCharges when prices or plans have changed. For tenure 0, fall back to the current MonthlyCharges (avoids divide-by-zero).
- **TenureGroup:** bucket tenure into customer-lifecycle stages. Churn risk is usually highest in the first year, so a group label can capture that pattern more directly than a raw month count. DECISION: bins of 0-12, 13-24, 25-48 and 49-72 months. The first two years get their own groups because that is where churn behavior changes fastest.

Churn rate by tenure group: new customers leave most



Encoding, splitting, and scaling

One-hot encoding

`encode_one_hot()` - Technique 2b: one-hot encode multi-category nominal columns.

> **DECISION:** one-hot rather than label encoding. These categories have no natural order ("Electronic check" is not "greater than" "Mailed check"), so label encoding them as 0/1/2/3 would invent a false ranking.

> **DECISION:** drop_first=True removes one dummy per column to avoid perfect multicollinearity, which matters for Logistic Regression.

Stratified 80/20 split

`split_data()` - Split into stratified 80/20 train and test sets.

> **DECISION:** split before scaling. The scaler must learn its mean/std from the training data only. If it were fit on the full dataset, information from the test set would leak into training and make model scores look better than they really are.

> **DECISION:** 80/20 split with stratify=y, which keeps the same churn ratio (~26.5%) in both sets. This matters because the classes are imbalanced.

Standard scaling

`scale_features()` - Technique 3: standardize continuous columns (fit on train only).

> **DECISION:** scale only the continuous columns and leave the 0/1 encoded columns as they are, so they stay easy to interpret.

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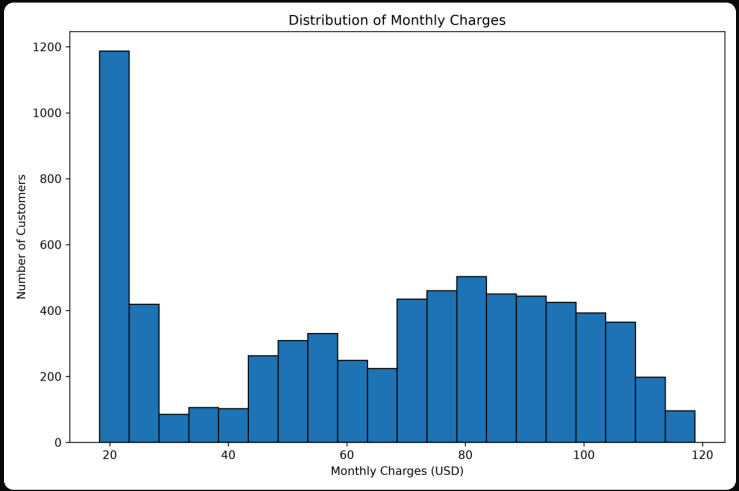
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EDA & Key Insights

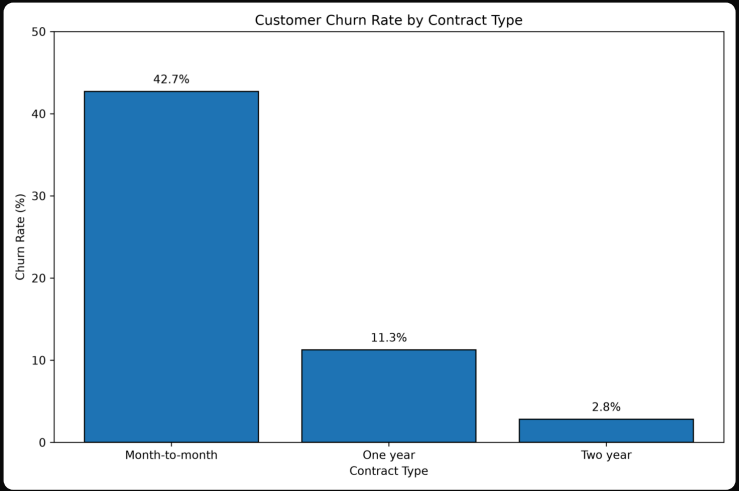
From `src/task_2_2_eda.py` , run on the cleaned data from Task 2.1.

Churn rate	Month-to-month churn	Median tenure (churned vs stayed)	Tenure-churn correlation
26.5%	42.7%	10 vs 38 months	-0.35

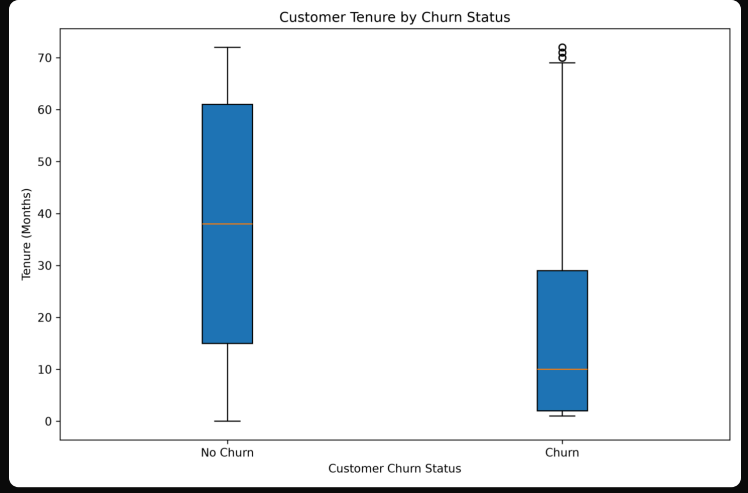
- **Contract type matters most:** month-to-month customers churn at 43%, against 3% on two-year contracts.
- **New customers leave:** churners have a median tenure of 10 months, stayers 38. Tenure has the strongest correlation with churn (-0.35).
- **Higher bills, more churn:** MonthlyCharges correlates positively with churn (0.19).
- **No outliers:** the IQR method finds no outliers in tenure, MonthlyCharges, or TotalCharges, so none were removed.



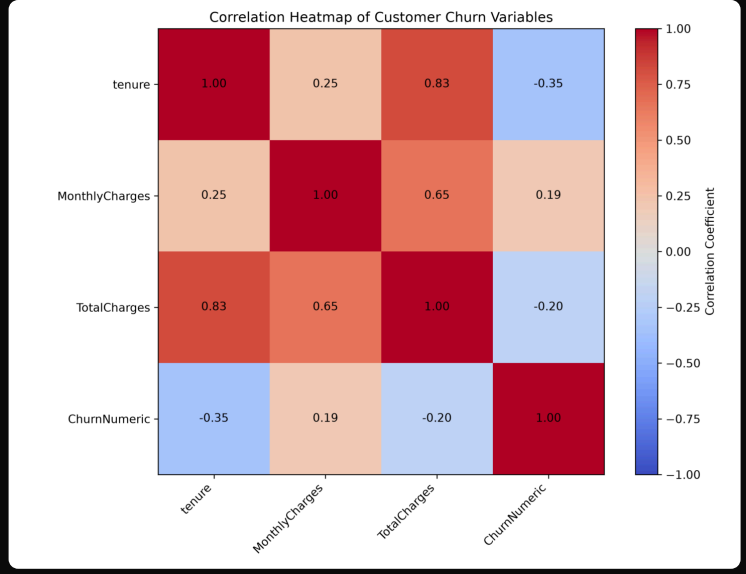
Distribution of monthly charges



Churn rate by contract type



Tenure by churn status



Correlation of numeric variables with churn

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Model Results

Two algorithms, **Logistic Regression** and **Random Forest**, both with `class_weight="balanced"` because only 26.5% of customers churn. Tuned with 5-fold `GridSearchCV` on the training set (scoring = F1). All numbers below are on the **held-out test set** (1,409 customers).

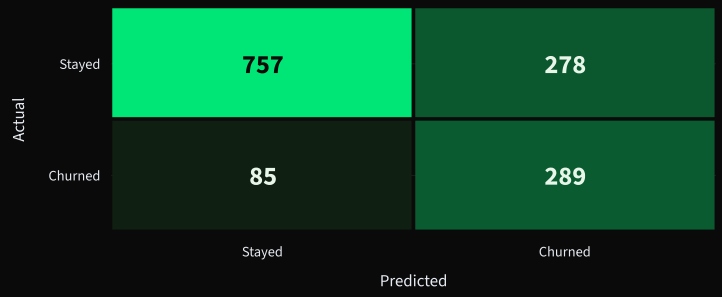
Comparison

Model	Accuracy	Precision	Recall	F1	ROC AUC
Baseline: predict everyone stays	0.735	0.000	0.000	0.000	0.500
Baseline Logistic Regression	0.733	0.498	**0.791**	0.612	**0.842**
Baseline Random Forest	**0.766**	**0.551**	0.636	0.591	0.826
Tuned Logistic Regression	0.742	0.510	0.773	0.614	0.839
Tuned Random Forest	0.757	0.528	0.773	**0.628**	**0.842**

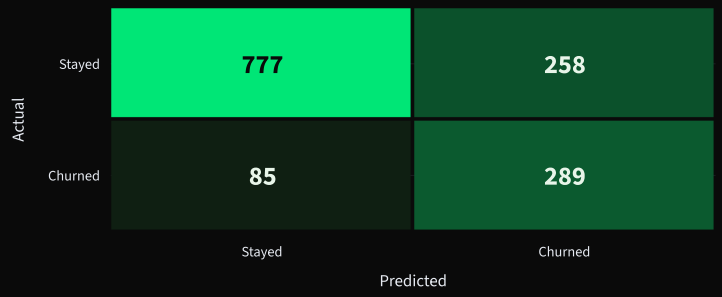
Green = best value in each column. The 'everyone stays' baseline is 73.5% accurate but catches no churners, which is why we judge the models on recall and F1, not accuracy.
Best parameters - Logistic Regression: {'C': 0.01}; Random Forest: {'max_depth': 10, 'n_estimators': 300}

Confusion matrices (threshold 0.5)

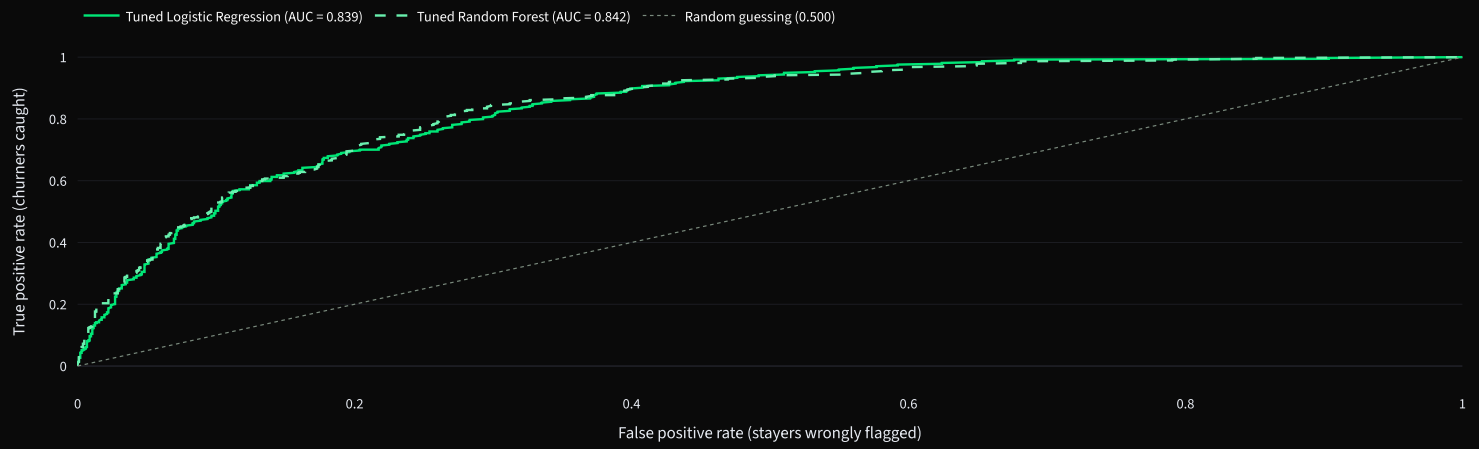
Tuned Logistic Regression



Tuned Random Forest



ROC curves



Choose the decision threshold

A model outputs a churn **probability**; the threshold decides when that counts as *will churn*. **Lower** the threshold and you catch more real churners (recall goes up), but you also flag more loyal customers by mistake (precision goes down). **Raise** it and the opposite happens. The right setting depends on what costs more: losing a customer, or sending a retention offer to someone who would have stayed.

Model

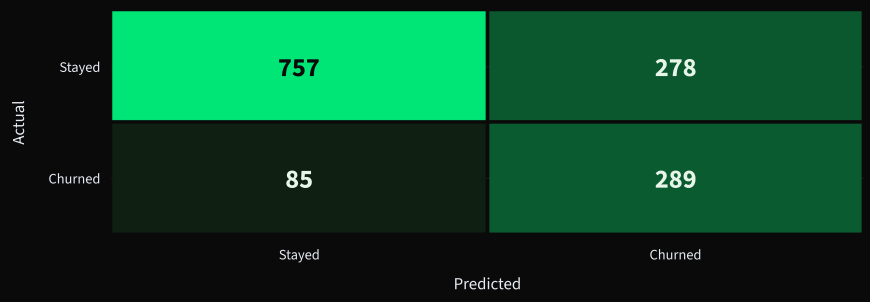
Tuned Logistic Regression

Threshold

0.50

Recall	Precision	Accuracy	F1
77.3%	51.0%	74.2%	61.4%

Tuned Logistic Regression at threshold 0.50



In plain English

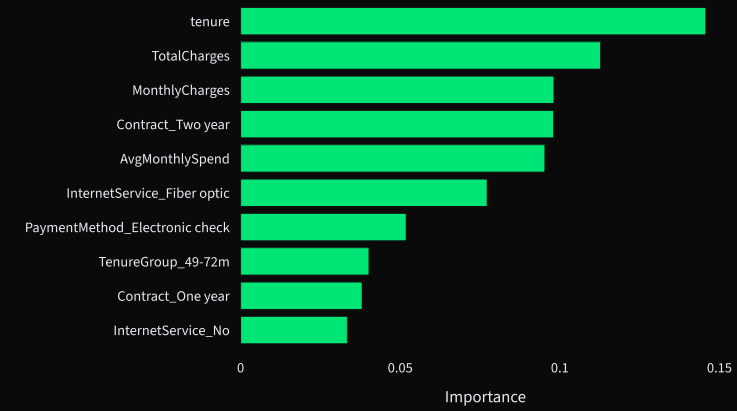
- Catches **289 of 374** customers who really left.
- Misses **85** churners (they leave without warning).
- Flags **278** customers who would have stayed (false alarms that would receive a retention offer).

What drives churn? Top 10 features

Logistic Regression coefficients
red = pushes toward Churned, green = pushes toward Stayed



Random Forest feature importances
how much each feature matters (no direction)



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Predict a Customer

Fill in a customer profile, or load an example, then press **Predict**.

Random customer | High risk example | Low risk example

Customer

Gender: Male

Senior citizen (65+): No

Has partner: Yes

Has dependents: Yes

Services

Phone service: Yes

Multiple lines: Yes

Internet service: DSL

Online security: Yes

Online backup: Yes

Device protection: Yes

Tech support: Yes

Streaming TV: No

Streaming movies: No

Account

Tenure (months): 60

Monthly charges (USD): 65.00

Contract: Two year

Paperless billing: No

Payment method: Credit card (automatic)

Total charges (tenure x monthly)

$3,900.00

Predict

Prediction

Logistic Regression

Stayed

↑ 5% churn risk

Random Forest

Stayed

↑ 4% churn risk

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Conclusion

Cleaning and Preprocessing

The raw dataset was already in good shape. It had no duplicate rows and no duplicate customer IDs. The only real problem was 11 blank TotalCharges values. All 11 belonged to customers with a tenure of 0 months, so we set them to $0. That is their true value, and it let us keep all 7,043 customers. We also changed "No internet service" and "No phone service" to "No". This avoided extra dummy columns, because InternetService and PhoneService already hold that information. We added three new features: NumServices, AvgMonthlySpend, and TenureGroup. We used one hot encoding with drop_first to avoid inventing a false order and to prevent multicollinearity. We split the data 80/20 with stratification before scaling, and we fit the scaler on the training set only. This kept test data out of training, so our scores reflect real performance. The final dataset had 28 features, with 5,634 training rows and 1,409 test rows.

Exploratory Data Analysis

The EDA showed that customer churn is strongly associated with contract type and customer tenure. Contract type showed one of the clearest differences in churn rates. Month to month customers churned at 42.7%, while customers on two year contracts churned at about 3%. Tenure was the strongest numeric predictor, with a correlation of −0.35. Customers who left had a median tenure of 10 months, compared with 38 months for those who stayed. The churn rate was approximately 47% among customers with 0-12 months of tenure and decreased to 10% among those with 49-72 months. MonthlyCharges had a weaker positive link with churn (0.19), so customers with higher bills left a bit more often. The IQR method found no outliers in tenure, MonthlyCharges, or TotalCharges, so we removed none. The overall churn rate of 26.5% also revealed a class imbalance, highlighting the importance of using evaluation metrics beyond accuracy. These findings suggest that newer customers and those on month-to-month contracts may deserve particular attention in customer retention strategies.

Model Training

We trained Logistic Regression and Random Forest with class_weight="balanced" to deal with the 26.5% churn rate. We tuned both models with 5 fold GridSearchCV on the training set, scored by F1, because accuracy alone hides poor churn detection. A model that predicts "everyone stays" scores 73.5% accuracy but catches zero churners. The best settings were C = 0.01 for Logistic Regression, and max_depth = 10 with 300 trees for Random Forest. Tuning helped Random Forest the most. Limiting tree depth cut overfitting, which raised recall from 63.6% to 77.3% and F1 from 0.591 to 0.628. Logistic Regression barely changed after tuning (F1 went from 0.612 to 0.614), which shows the default model was already close to its best.

Model Comparison Results

The tuned Random Forest performed best overall. It had the highest F1 (0.628) and a slightly higher ROC AUC (0.842) than Logistic Regression (0.839). The tuned Logistic Regression came very close, with an F1 of 0.614 and an AUC of 0.839. Both tuned models caught 289 of the 374 churners in the test set, a recall of 77.3%. The difference was in false alarms. Random Forest wrongly flagged 258 loyal customers, and Logistic Regression flagged 278. This gave Random Forest slightly better precision (52.8% vs 51.0%) and accuracy (75.7% vs 74.2%). Both models agreed on the main drivers of churn: short tenure, month to month contracts, fiber optic internet, and electronic check payments. Logistic Regression adds direction to these findings. Two year contracts and longer tenure push customers toward staying, while electronic checks and fiber optic push them toward leaving.

Overall Conclusion

This project shows that we can predict customer churn well with standard, interpretable models. The tuned Random Forest is our recommended model because it has the best F1 and AUC. The gap between the two models is small, though. If the business needs to explain to managers why a customer is at risk, **Logistic Regression** is a strong choice because its coefficients show which way each feature pushes. Both models find about three out of four customers who will leave. The cost is that roughly half of the flagged customers would have stayed anyway. The company can adjust the decision threshold depending on whether losing a customer costs more than sending an unneeded retention offer. The results point to clear actions. The company should focus retention efforts on customers in their first year, move month to month customers onto longer contracts, and look into why fiber optic and electronic check customers leave more often. Next steps could include testing other algorithms such as gradient boosting and adding cost data to choose the best threshold.

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References

1. BlastChar. (n.d.). *Telco customer churn* [Data set]. Kaggle. Telco Customer Churn | Kaggle